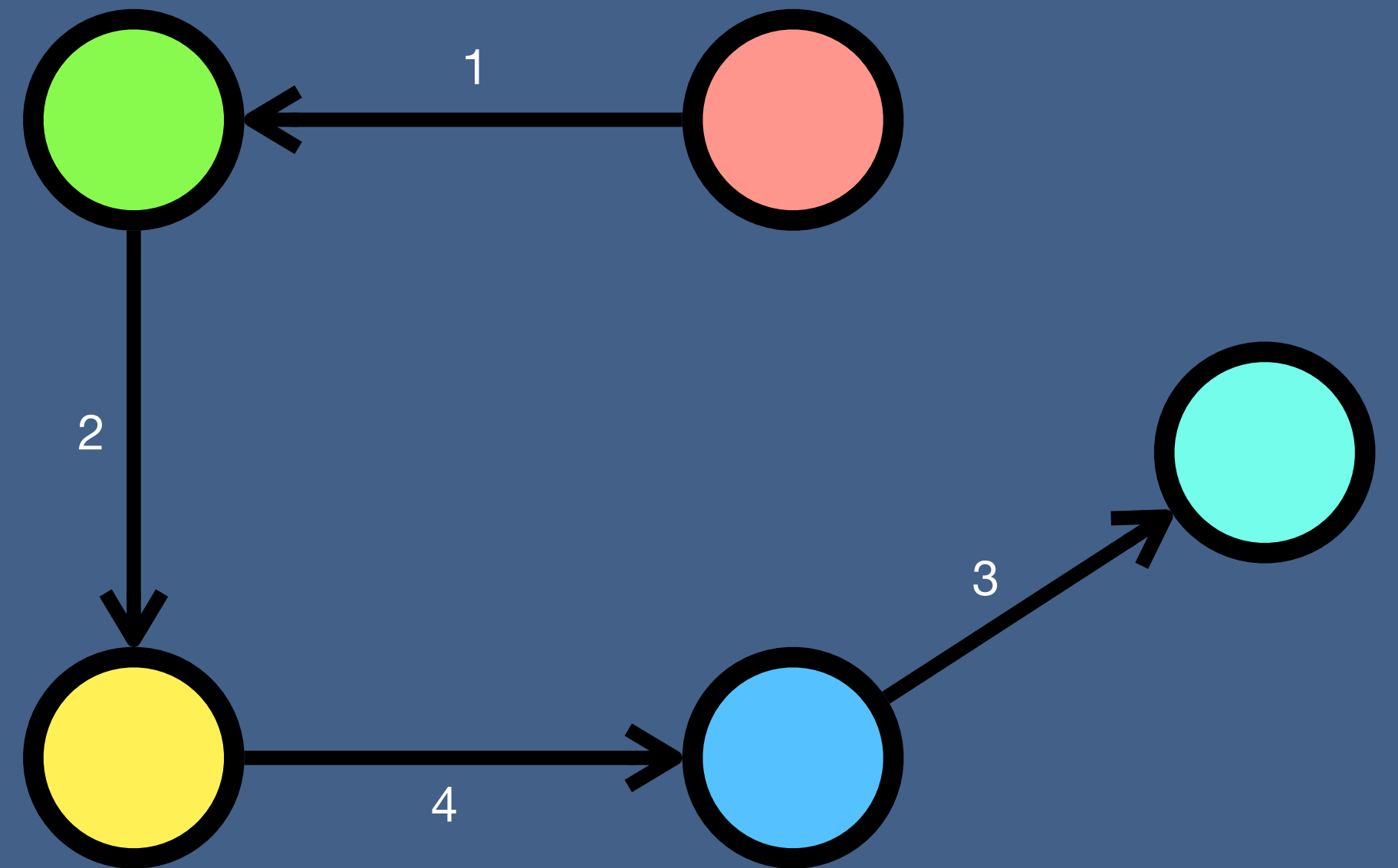


Kruskal's Algorithm

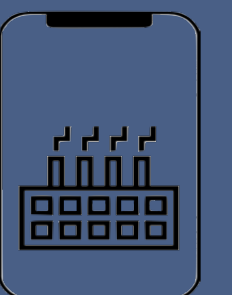
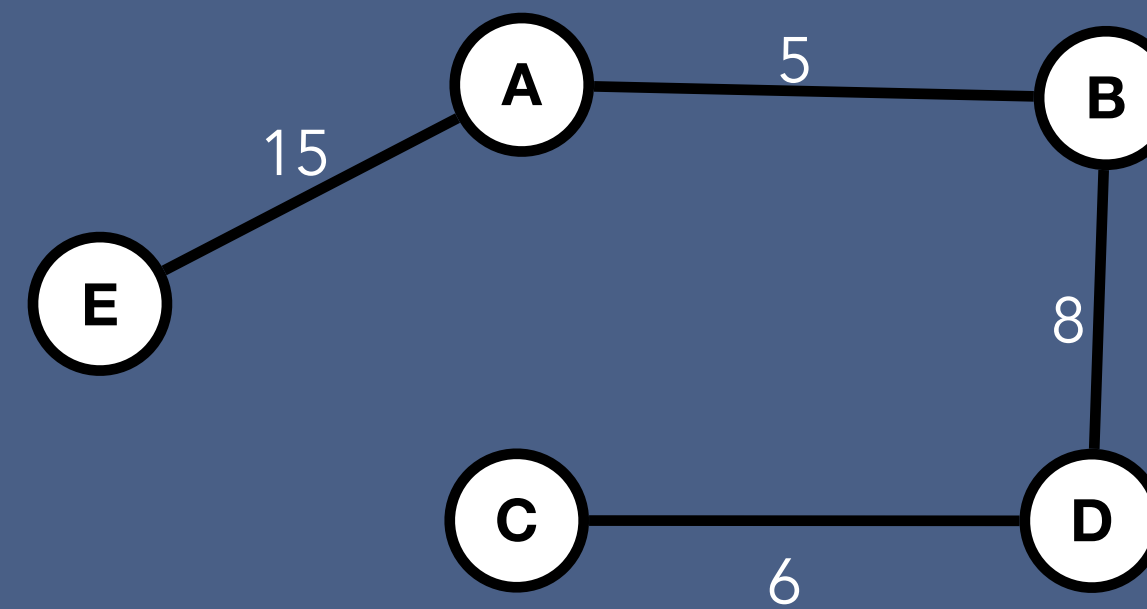
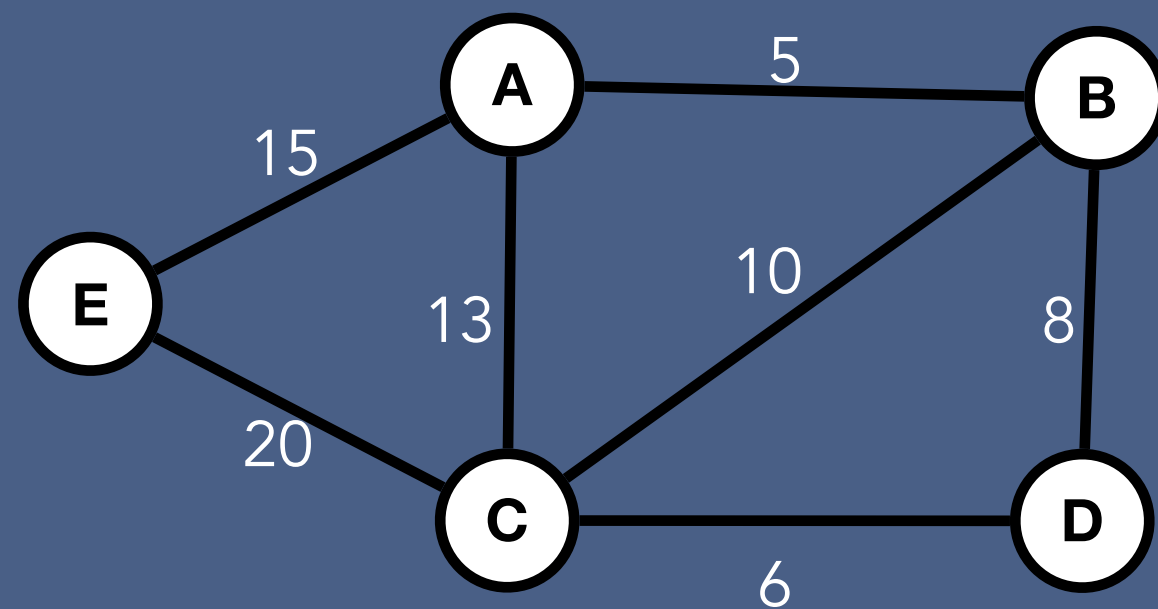


Kruskal's Algorithm

It is a greedy algorithm

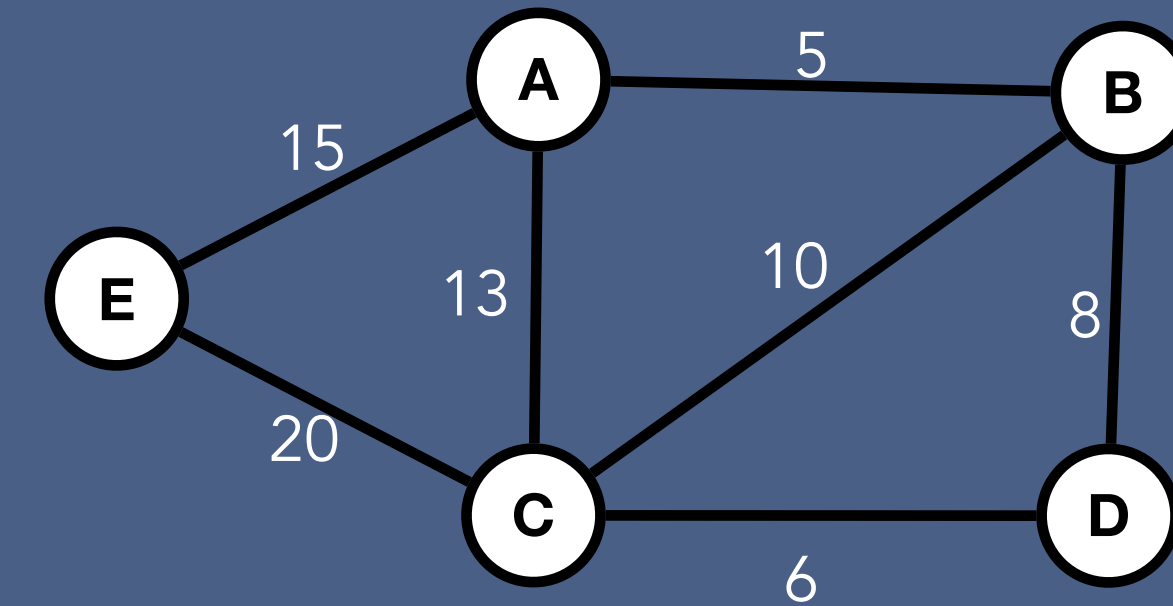
It finds a minimum spanning tree for weighted undirected graphs in two steps

- Add increasing cost edges at each step
- Avoid any cycle at each step

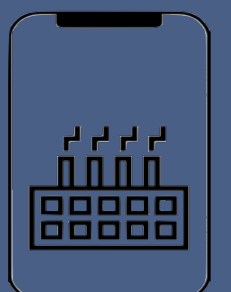
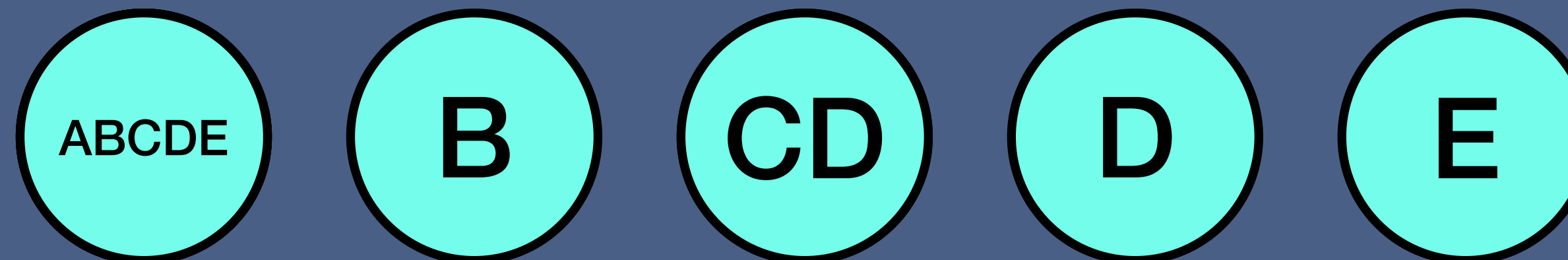
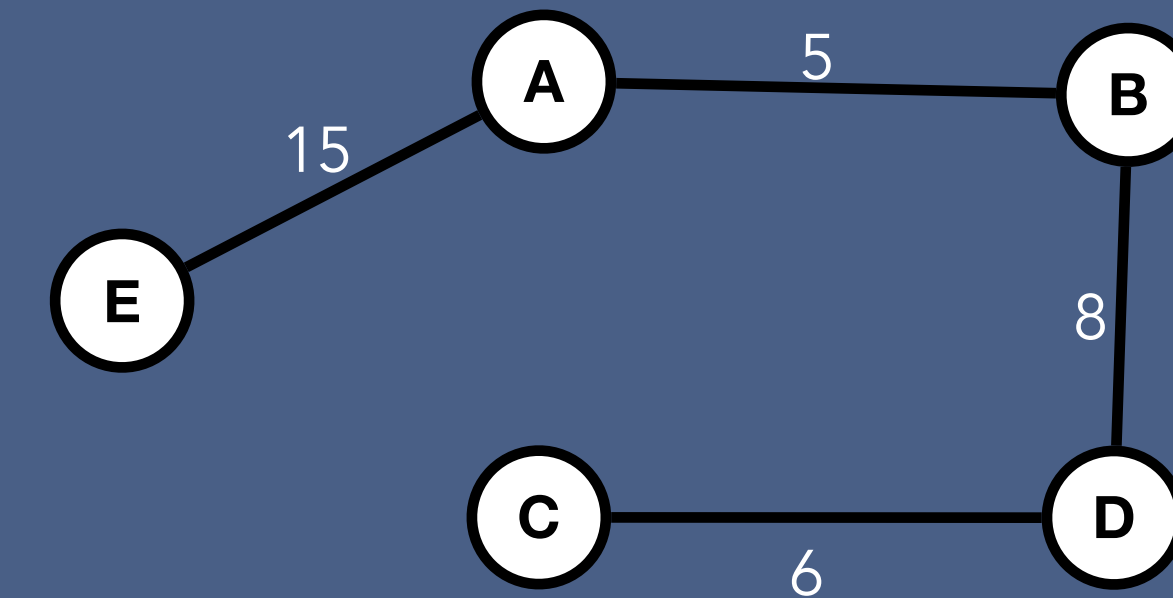


Kruskal's Algorithm Pseudocode

```
Kruskal(G)
for each vertex
    makeSet(v)
sort each edge in non decreasing order by weight
for each edge (u, v)
    if findSet(u) ≠ findSet(v)
        union(u, v)
        cost = cost + edge(u,v)
```



Cost = 0 + 5 + 6 + 8 + 15 = 34



Time and Space Complexity of Kruskal's Algorithm

```
Kruskal(G)
for each vertex .....→ O(v)
  makeSet(v)
sort each edge in non decreasing order by weight .....→ O(elog e)
for each edge (u, v) .....→ O(e)
  if findSet(u) ≠ findSet(v) .....→ O(1)
    union(u, v) .....→ O(v)
    cost = cost + edge(u,v).....→ O(1) } .....→ O(ev)
```

Time complexity : $O(V + E \log E + EV) = O(E \log(E))$

Space complexity : $O(V + E)$

